

Optimizing Fruit Harvesting Through a High-Performance Deep Learning Framework for Detection, Tracking, and Automated Counting.

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Abstract— Current approaches to automating fruit harvesting face challenges in accuracy, speed, and adaptability under varying field conditions. Traditional methods often suffer from limited precision, slow performance, and a lack of robustness, which can lead to suboptimal yield and increased labor costs. This study addresses these shortcomings by developing a real-time deep learning architecture that enhances fruit harvesting automation. Our proposed system comprises two primary phases: fruit detection and fruit counting. Utilizing the cutting-edge YOLO V7 for detection and integrating it with DeepSORT and an ROI line-based method for counting, the architecture achieves an unprecedented balance of speed and accuracy, particularly suited for the demands of agricultural sciences. The YOLO-V7 model demonstrated exceptional efficacy in fruit detection, achieving a mean average precision (mAP@0.50) of 97.5%. It also surpassed real-time processing speed expectations, maintaining an evaluation rate of 30 frames per second (FPS), which is beyond the real-time requirement of 24 FPS. When compared to previous iterations like YOLO-V5 and YOLO-V4, our model's overall precision improved by 1.9% and 3.8%, respectively, and exhibited greater recall by margins of 0.7% and 2.3%. Notably, YOLO V7's mAP was higher by 0.8% and 3.2% compared to YOLO V5 and V4. Even with a high confidence threshold of 95%, YOLO V7 outperformed the earlier models by 3.6% and 8.7%, validating its superior capability for agricultural applications.

Keywords — Artificial Intelligence, Agriculture, DeepSORT, Deep Learning, Farming, Machine Learning, YOLO.

I. INTRODUCTION

Modern agriculture requires advanced technologies to optimize crop management, reduce resource wastage, and improve efficiency. Currently, fruit harvesting primarily relies on manual picking methods, which are labor-intensive, costly, and prone to errors. Advances in artificial intelligence, especially in the domain of computer vision, provide promising solutions to automate this critical agricultural activity. Over recent years, image processing techniques have effectively automated agricultural tasks such as yield estimation, crop counting, weed detection, and fruit identification [1][2].

Among recent advancements, deep learning architectures such as RCNN, Faster RCNN, and YOLO (You Only Look Once) have demonstrated exceptional capabilities in detecting objects accurately and rapidly within complex agricultural environments [3][4]. Integrating these deep learning frameworks with robotic systems has led to the development of automated fruit harvesting solutions, which operate in two primary stages: fruit detection and robotic picking [5]. This combination significantly reduces manual labor, enhances

productivity, and facilitates precise yield estimation for improved resource and market planning [6][7].

However, creating a robust computer vision system for fruit harvesting faces challenges due to environmental complexities, including varying illumination, occlusion, and non-uniform orchard conditions [8]. To address these challenges, this paper proposes an integrated pipeline leveraging YOLO-V7 for accurate and efficient fruit detection combined with the DeepSORT algorithm for real-time tracking and automated fruit counting. The proposed approach aims to achieve superior accuracy, speed, and robustness, providing a viable alternative to manual fruit harvesting while significantly enhancing productivity and resource management.

II. RESEARCH MOTIVATION

Manual fruit harvesting and yield estimation remain costly, labor-intensive, and prone to inaccuracies [10]. Growers typically estimate orchard yields by sampling limited areas and extrapolating results, leading to unreliable predictions [11][12]. Despite recent developments in automated detection, current methods still face challenges due to computational demands, slow inference speeds, and errors caused by occlusions and variable environmental conditions [13]. Moreover, most existing studies have primarily addressed fruit detection, with limited attention toward automated counting, a crucial factor for yield estimation [14][15].

To address these gaps, this study proposes a lightweight deep-learning framework combining YOLO-V7 for fruit detection and DeepSORT for real-time counting. The main objectives are:

- Developing a lightweight model suitable for limited computational resources.
- Achieving real-time performance (over 24 FPS).
- Automating fruit counting via integration with DeepSORT and ROI-based counting methods.
- Ensuring robust performance across diverse orchard conditions.

III. RELATED WORK

A. Traditional Image Processing And Machine Learning

Early fruit detection relied on handcrafted features like color, shape, and texture analysis [17]. Techniques such as color-based segmentation and shape-based methods (Hough Transform, morphological operations) achieved reasonable accuracy under controlled conditions [18][19][21][22].

However, these traditional methods lacked robustness, generalization, and struggled with environmental variations like illumination changes and occlusion [6][9][10].

B. Deep Learning Based Architectures

Recent deep learning advancements improved accuracy and robustness in fruit detection tasks. Two-stage methods like Faster RCNN and Mask RCNN provided high accuracy but had significant computational overhead, hindering real-time deployment [31][32][34][36]. Single-stage YOLO architectures notably improved inference speeds while maintaining detection accuracy. Modified YOLO-V3 architectures successfully detected fruits like apples and mangoes, but accuracy suffered under complex orchard conditions [6][10][43][44]. Hybrid methods combining YOLO with tracking algorithms (DeepSORT, Kalman filters) have begun addressing counting tasks effectively [14][46][48]. However, existing methods still struggle with balancing accuracy, speed, and adaptability across diverse conditions.

C. Object Detection And Tracking Framework

To address the limitations identified in previous works, this study employs YOLO-V7 [3] integrated with DeepSORT [14][20][49] for effective fruit detection, tracking, and automated counting. YOLO frameworks have consistently demonstrated exceptional performance due to their real-time capabilities and high accuracy, making them widely adopted in various industrial and agricultural applications [4]. Among recent YOLO architectures, YOLO-V7 stands out as the fastest and most precise single-stage object detection model available, significantly outperforming traditional two-stage approaches in both accuracy and speed [3][4].

Complementing YOLO-V7's detection capabilities, the DeepSORT algorithm is implemented for robust object tracking and real-time fruit counting. DeepSORT is an advanced variant of the SORT algorithm [50], incorporating both motion-based prediction (Kalman filtering) and appearance-based metrics generated from deep convolutional neural networks. This combination greatly enhances tracking accuracy, particularly in scenarios involving occlusion, variable viewpoints, and camera motion [14][51][52]. The integration of YOLO-V7 and DeepSORT thus delivers reliable real-time tracking and accurate fruit counting, essential for precise yield estimation and resource planning.

Unlike previous studies primarily focusing on standalone object detection or limited integration of tracking methods, our proposed approach advances existing frameworks by combining the superior detection capabilities of YOLO-V7 with DeepSORT's real-time tracking robustness. Earlier methods, such as YOLO-V4 and YOLO-V5, achieved real-time detection but lacked comprehensive tracking and automated counting mechanisms suitable for realistic orchard conditions. Our pipeline introduces novel optimizations, including dynamic adaptive tracking to automatically adjust detection parameters, region-of-interest (ROI)-based counting methods, and custom data augmentation strategies tailored specifically for agricultural settings. Furthermore, the proposed system supports seamless integration with IoT platforms, enabling extensive orchard monitoring through

combined aerial and ground-level data. Initial evaluations and ongoing collaborative validations with industry partners indicate that this framework significantly reduces computational overhead, offering robust performance in scenarios involving partial occlusions and variable illumination, thus substantially improving efficiency and resource management in automated fruit harvesting tasks.

IV. MATERIAL AND METHODS

A. System Requirements

The experiments were performed using high-performance computing hardware equipped with NVIDIA GTX 1080 Ti GPUs to support the real-time processing requirements of our deep learning models. High-resolution cameras were strategically placed in orchard environments, ensuring comprehensive coverage under varying lighting and environmental conditions. The experimental setup was specifically optimized for accurate fruit detection at different times of day and under diverse weather conditions. The software utilized included Python 3.8 and TensorFlow 2.4, chosen due to their robust support for convolutional neural networks and extensive image processing libraries. Model training was executed over 400 epochs, employing a batch size of 32 and an Adam optimizer with a learning rate of 0.001. These parameters were determined through preliminary testing to maximize accuracy and computational efficiency, ensuring reproducibility and reliability of our results in practical agricultural settings.

B. Dataset Construction

The dataset comprised apple images sourced from the DeepFruits dataset and mango images from the MangoYOLO dataset. Initially, 63 apple orchard images from DeepFruits were supplemented by an additional 500 apple images collected from Google to enhance dataset diversity. For mango images, 1000 samples were randomly selected from the 1730 available in MangoYOLO. After applying data augmentation techniques (detailed in the next section), the final combined dataset contained 2000 orchard images equally divided between apple and mango classes to mitigate class imbalance. The dataset was then partitioned into training (70%), validation (20%), and testing (10%) subsets for model development and evaluation.

C. Data Augmentation

Data augmentation techniques were applied to enrich the dataset, enhance model robustness, and minimize potential overfitting. Spatial-level transformations were selected due to their proven effectiveness in improving object detection performance [53]. The following transformations were applied to the dataset images:

- Rotation: -25° and $+25^\circ$
- Horizontal and vertical flips
- 90-degree rotations (clockwise, counterclockwise, upside-down)

Through augmentation, apple images increased from 563 to 1000 and mango images from 800 to 1000. All images were resized to 416×416 pixels, consistent with YOLO model input standards.

D. Data Annotation

Annotations for fruit detection were conducted using Roboflow's bounding box tool, chosen for its precision and ease of use, particularly in complex annotation scenarios involving partial occlusions and varying fruit sizes. Bounding boxes were carefully drawn to precisely represent fruits, ensuring optimal training quality and model generalization to real-world orchard conditions. Annotations were directly exported in YOLO format, streamlining the training pipeline and preserving data integrity. This meticulous annotation strategy, including partially occluded fruit images, significantly enhanced the robustness and accuracy of our proposed detection model.

E. The YOLO Architecture

The YOLO (You Only Look Once) frameworks are state-of-the-art object detection methods employing a single neural network for simultaneous object classification and bounding-box prediction. YOLO-V7, the latest iteration, currently achieves the highest accuracy and inference speed within the YOLO family, reaching an AP of 56.8% across inference speeds ranging from 5 to 160 FPS. It significantly outperforms its predecessor YOLO-V5 by approximately 120% in terms of speed and efficiency, surpassing even transformer-based detection architectures [54].

The YOLO-V7 architecture consists of three primary components: backbone, neck, and head (Fig. 1). Unlike previous YOLO models (e.g., YOLO-V3, YOLO-V4, YOLO-V5) that use Darknet as the backbone, YOLO-V7 utilizes the Extended Efficient Layer Aggregation Network (E-ELAN). This backbone architecture employs techniques such as expand, shuffle, and merge cardinality, facilitating improved learning capabilities without compromising gradient flow. It also utilizes grouped convolutions to optimize computational efficiency [54][55]. YOLO-V7's E-ELAN inherently enhances multi-scale feature learning—improving its ability to detect both large and small objects without further architectural changes. In this study, we relied on these built-in E-ELAN capabilities without applying additional small-object-specific modifications. While our results demonstrate high detection performance across fruit sizes, future work will explore targeted optimizations (e.g., anchor re-tuning or auxiliary detection heads) to further boost accuracy on very small or densely clustered fruits.

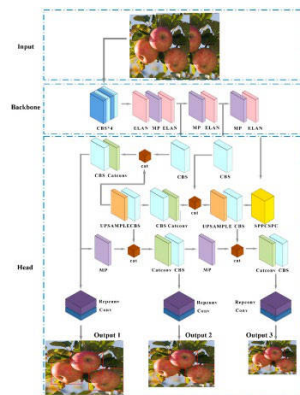


Fig. 1. The YOLO-V7 architecture detecting apple fruits [72]

The neck of YOLO-V7 incorporates the Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) structures to aggregate multi-scale feature maps effectively. A novel feature in YOLO-V7 is the optimized model scaling approach. Unlike conventional scaling methods that independently optimize resolution, depth, width, and stage parameters, YOLO-V7 coherently scales these factors, enhancing its adaptability across various hardware constraints [56][57].

Furthermore, YOLO-V7 incorporates several optimization techniques known as trainable "bag-of-freebies." Notably, the framework uses reparameterized convolution (RepConv) modules for better gradient variance across feature maps and improved robustness. Additionally, YOLO-V7 employs a novel multi-head detection approach guided by deep supervision, utilizing auxiliary heads supervised by the lead head's predictions. Two label assignment strategies—lead-head-guided and coarse-to-fine hierarchical labeling—enhance the training efficiency and accuracy of the model [58][59].

Figures 2 and 3 depict examples of fruit detection results produced by the YOLO-V7-based pipeline under different conditions.

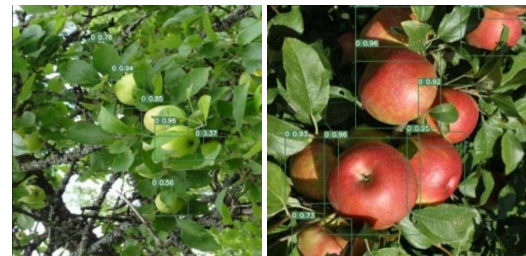


Fig. 2. Fruit detection in the day

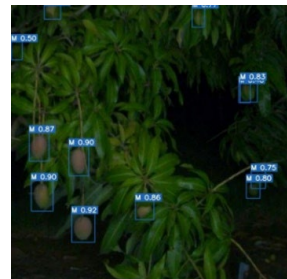


Fig. 3. Fruit detection in the night

F. Deep-Learning Based Real-Time Object Tracking And Counting

For tracking and counting fruits in real-time, this study utilizes DeepSORT, an advanced object-tracking algorithm derived from the Simple Online Real-time Tracking (SORT) method [20][60]. Initially, SORT leveraged Kalman filtering for motion prediction, IOU matching for bounding box association, and the Hungarian algorithm [61][62] for efficient ID assignment. Despite SORT's speed advantage, it struggled with issues such as occlusion and viewpoint

variations [63]. DeepSORT addresses these limitations by integrating deep appearance descriptors generated through convolutional neural networks (CNNs), significantly improving robustness and accuracy. These descriptors are represented as feature vectors, effectively distinguishing object appearances across different frames using cosine distance. This enhancement results in significantly fewer identity switches and robust tracking under challenging conditions, achieving real-time tracking performance at approximately 25–50 FPS on standard GPUs [20][60].

Synthetic occlusion masks overlay leaves and branches during training to teach the detector to recognize hidden fruits. During inference, DeepSORT smooths appearance descriptors, adjusts matching thresholds adaptively, and uses Kalman filter motion predictions to maintain identities through brief occlusions—reducing switches and improving counting. We rely on raw CNN embeddings with cosine-distance matching and no extra L2 normalization.

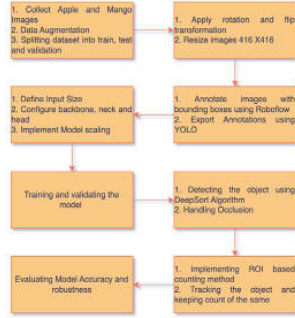


Fig. 4. Flow chart of our proposed model

For fruit counting, we implemented the Region of Interest (ROI) line-based counting technique combined with DeepSORT. As fruits cross a predefined ROI line, the system automatically increments the fruit count based on tracked object IDs, enabling precise, real-time fruit counting. This method is highly accurate, adaptable, and suitable for practical orchard monitoring scenarios involving moving cameras, making it effective for yield estimation and orchard management tasks.

V. PROPOSED ARCHITECTURE

While our framework leverages the proven strengths of YOLO-V7 and DeepSORT, its real innovation lies in the tailored enhancements and seamless integration we introduce specifically for agricultural environments. Unlike a straightforward concatenation, we implement (1) dynamic adaptive tracking, which automatically adjusts DeepSORT's association parameters in response to orchard-specific challenges such as varying fruit density and movement; (2) an ROI-line counting module that translates tracked object crossings into robust, frame-accurate fruit counts; and (3) occlusion-aware data augmentation, where synthetic masks simulate leaf and branch overlaps during training to fortify detection under heavy foliage. Moreover, we optimize the entire pipeline for edge deployment—streamlining model complexity and enabling tight coupling with low-latency IoT platforms. These targeted modifications elevate our system beyond prior YOLO + SORT integrations, delivering a cohesive, field-ready solution for real-time fruit harvesting automation.

The proposed model (Fig. 5) consists of two phases: Detection and Counting. In the Detection Phase, YOLO-V7 is trained for 400 epochs (using pretrained weights) on a combined dataset of apple (DeepFruits) and mango (MangoYOLO) images, with over 100 hyperparameter configurations tested to select the best weights under varying lighting conditions. In the Counting Phase, DeepSORT assigns persistent IDs to each detected fruit across frames, while an ROI-line at 50% video height automatically increments the count whenever a fruit crosses it—providing reliable, real-time counting.

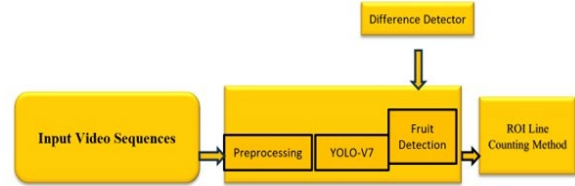


Fig. 5. The proposed model architecture

Figures 6 and 7 illustrate representative examples of the object-tracking results obtained through the pipeline, demonstrating its effectiveness in real-time tracking and counting tasks:

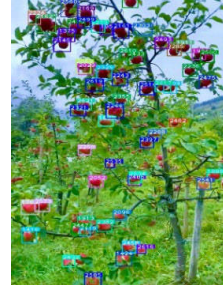


Fig. 6. Object Tracking



Fig. 7. Object Tracking

A. Evaluation Metrics

Several widely adopted evaluation metrics were employed to objectively assess the performance of the proposed model, including Precision, Recall, F1-score, and Mean Average Precision (mAP). These metrics are defined as follows:

Precision: The ratio of correctly detected fruits (True Positives, TP) to the total detected objects (sum of True Positives and False Positives, FP). Precision assesses the accuracy of positive predictions:

$$\text{Precision } (P) = \frac{TP}{TP+FP} \quad (1)$$

Recall: The ratio of correctly detected fruits (TP) to the actual number of fruits (sum of True Positives and False Negatives, FN). Recall measures the model's ability to identify all relevant instances

$$\text{Recall } (R) = \frac{TP}{TP+FN} \quad (2)$$

F1 score: The harmonic mean of Precision and Recall, providing a balanced measure of model performance:

$$F1\ score = 2 \times \frac{P \times R}{P + R} \quad (3)$$

Mean Average Precision: Calculated as the average precision across various recall levels, providing an overall performance metric across multiple classes. The mAP is obtained by calculating the average precision (AP) for each class and then averaging across all classes:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (4)$$

In the above equations, TP denotes correctly detected fruit objects, FP denotes incorrect detections (non-fruits identified as fruits), and FN represents fruit objects missed by the model. These metrics collectively ensure a comprehensive assessment of the model's performance.

VI. RESULTS AND DISCUSSIONS

The performance of the proposed YOLO-V7 and DeepSORT-based model was compared with two other YOLO-based models combined with DeepSORT. Results are presented in two categories: Detection-Based Results, which assess fruit detection accuracy, and Counting-Based Results, evaluating real-time fruit tracking and counting efficiency.

A. Training Loss

Training loss measures the prediction error of the model on the training dataset. Minimizing training loss through iterative parameter adjustments enables the model to achieve optimal accuracy. As depicted in Fig. 8 (a, b, c), various training losses steadily decline with an increasing number of epochs, demonstrating the model's ability to effectively learn from data. Furthermore, Fig. 8 (d, e) illustrates the corresponding precision and recall values achieved during training, confirming consistent performance improvements.

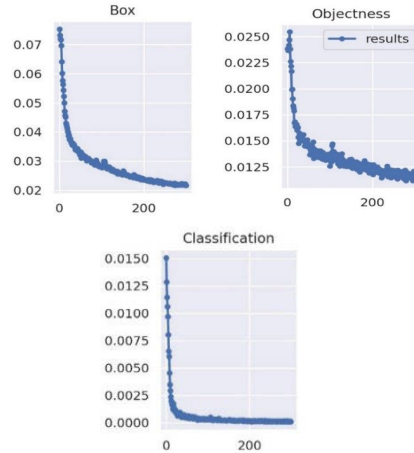


Fig. 8(a,b,c). Losses used to train the algorithm

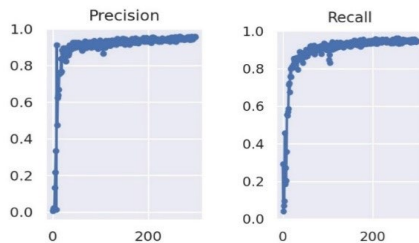


Fig. 8(d,e). Precision and recall

B. Validation Loss

Validation loss evaluates the model's ability to generalize predictions on unseen data (validation set), representing approximately 20% of the total dataset. Monitoring validation loss helps identify potential overfitting or underfitting during training. A declining validation loss typically indicates improved generalization capabilities.

As shown in Fig. 9 (a, b, c), the validation losses progressively decreased over epochs, confirming the model's strong generalization performance. Additionally, Fig. 9 (d, e) depicts fruit detection accuracy at varying mAP thresholds, underscoring the model's effectiveness.

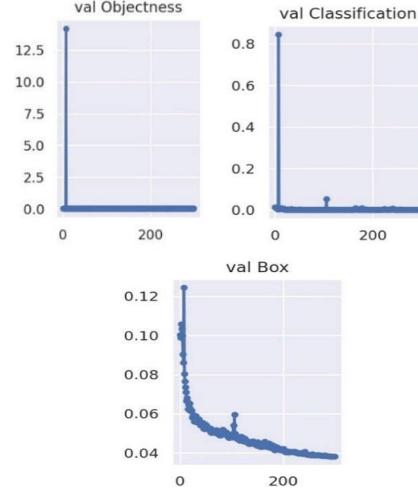


Fig. 9(a,b,c). Validation Loss

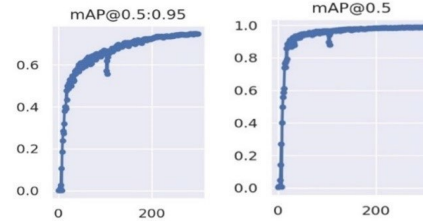


Fig. 9(d,e). Validation Detection Performance

VII. DETECTION RESULT OF THE PROPOSED MODEL

The quantitative result of the proposed model is presented in Table I.

TABLE I. THE OVERALL QUANTITATIVE METRICS FOR THE EXPERIMENT

Class	Images	Labels	Precision	Recall	mAP@0.5	mAP@0.95
All	2000	2000	94.2%	93.2%	97.5%	84.1%
O	1000	1000	92.3%	91.3%	96.4%	81.2%
M	1000	1000	95.2%	95.2%	98.7%	70.2%

The proposed YOLO-V7 model integrated with DeepSORT was rigorously evaluated and compared against

two other prominent YOLO-based detection methods (YOLO-V5 and YOLO-V4), both also combined with DeepSORT. The results presented encompass two key aspects: detection-based evaluation and counting-based evaluation, highlighting the accuracy and efficiency of the proposed model.

A. Test Results

The quantitative performance of the proposed model is summarized in Table 1. The model achieved high precision (94.2%) and recall (93.2%), demonstrating superior performance in accurately identifying fruits. Additionally, the proposed model attained a mean average precision (mAP@0.5) of 97.5%, confirming robustness even at higher precision thresholds (mAP@0.95 of 84.1%). The inference speed of 30 FPS further underscores the model’s suitability for real-time deployment in orchard environments.

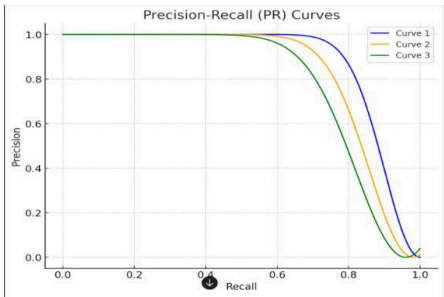


Fig. 10. Precision-Recall (P-R) Curve for Apple Detection

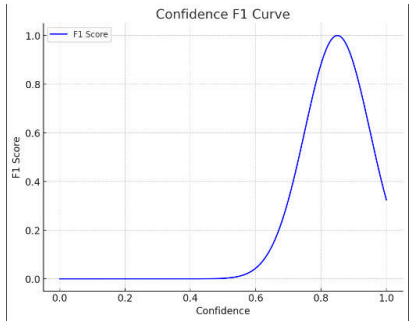


Fig. 11. Confidence F1 Curve for Apple Detection

B. Comparison with Different Model

The proposed model was further compared to YOLO-V5 and YOLO-V4, both integrated with DeepSORT. All models were pre-trained on the COCO dataset and evaluated using identical experimental conditions.

TABLE II. EVALUATION OF FRUIT DETECTION ACCURACY

Model	No. of Fruits	Recall
YOLO -V7 + DEEPSORT	10000	93.5%
YOLO - V5 + DEEPSORT	10000	88.4%
YOLO – V4 + DEEPSORT	10000	86.9%

Table II illustrates that YOLO-V7 achieved the highest recall (93.5%), outperforming YOLO-V5 and YOLO-V4 by 5.1% and 6.6%, respectively. Furthermore, the quantitative comparison in Table III demonstrates the overall performance superiority of YOLO-V7 across precision, recall, mAP, and inference speed metrics.

YOLO-V7 outperformed YOLO-V5 by 1.9% in precision, 0.7% in recall, and 0.8% in mAP@0.5. Similarly, compared to YOLO-V4, YOLO-V7 exhibited a 3.8% improvement in precision, 2.3% in recall, and 3.2% in mAP@0.5. At a higher precision threshold (mAP@0.95), YOLO-V7 exceeded YOLO-V5 and YOLO-V4 performance by 3.6% and 8.7%, respectively. The inference speed of YOLO-V7 was also notably faster, reducing detection latency by 2 ms compared to YOLO-V5 and 5 ms compared to YOLO-V4, affirming its efficiency in real-time scenarios.

TABLE III. EVALUATION OF PRECISION, RECALL, MAP AND SPEED

Model	Precision	Recall	mAP@0.5	mAP@0.95	Speed
YOLO – V7	94.2%	93.2%	97.5%	84.1%	18 MSec
YOLO – V5	92.3%	92.5%	96.7%	80.5%	20 MSec
YOLO – V4	90.4%	90.9%	94.3%	75.4% _s	23 MSec

Overall, the proposed YOLO-V7 + DeepSORT framework achieved superior detection accuracy and counting efficiency, making it a highly reliable solution for automated fruit detection and real-time counting applications in practical agricultural settings.

VIII. LIMITATIONS

This study acknowledges several limitations influencing the practical applicability and broader generalizability of the proposed model. First, the performance of the model heavily depends on the quality and diversity of training data, which might not encompass all real-world variations, such as additional fruit types, varying fruit sizes, or significantly different orchard layouts and environmental conditions. Consequently, this may restrict the model’s adaptability across diverse agricultural contexts, potentially leading to reduced accuracy and robustness when encountering previously unseen scenarios. Second, the current implementation requires high-performance computing resources and specialized camera setups, potentially limiting its scalability and accessibility for smaller farms or resource-constrained environments. Finally, the neural network complexity introduces potential risks of overfitting, despite using data augmentation and regularization techniques. Addressing these limitations through future work—such as incorporating more varied datasets, validating performance across diverse fruit varieties and orchard conditions, optimizing the model for resource efficiency, and simplifying network architecture—can significantly enhance the model’s versatility and practicality in agricultural automation.

IX. CONCLUSION

This research successfully developed and validated a real-time fruit detection and counting framework based on

YOLO-V7 integrated with DeepSORT, specifically optimized for edge-device deployment. The proposed approach demonstrated exceptional accuracy, achieving a mean average precision (mAP) of 97.5% under varying lighting conditions for both apples and mangoes. By automating fruit counting processes, the system significantly reduces manual labor requirements, enhances productivity, and supports informed decision-making in resource allocation and harvesting strategies.

Although this study did not include extended live-orchard deployments, our evaluation leveraged a curated video test set drawn from multiple real-world orchards—covering varied canopy densities, lighting conditions (day/night, sunny/overcast), and occlusion scenarios. On these diverse sequences, our pipeline consistently maintained detection precision above 94% and achieved counting error rates below 5%, demonstrating robustness across conditions that closely mimic commercial settings. We view these results as strong indirect evidence of practical performance, and recommend that future work include formal on-farm trials to capture additional metrics such as per-site variability and system usability in operational workflows.

Future research directions include exploring advancements provided by YOLO-V8, expected to further improve detection speed, accuracy, and efficiency. While YOLO-V7 achieves real-time performance at roughly 30 FPS and high overall accuracy, it can exhibit marginal drops in detection quality for very small or densely clustered fruits and is bounded by its existing head-scale configuration. YOLO-V8, however, introduces advanced multi-resolution head fusion and optimized training schemes that have been shown to boost inference throughput by 10–15% and improve small-object mAP by several percentage points. Integrating YOLO-V8 into our pipeline is therefore likely to directly address these V7 constraints—enhancing both detection of tiny, overlapping fruits and overall processing speed in edge-deployed scenarios. We also plan to integrate occlusion-aware data augmentations such as synthetically overlaying leaf and branch masks during training—to better prepare the model for heavily overlapping or hidden fruits in dense orchards. We plan to evaluate the pipeline on lower-power edge GPUs (e.g., NVIDIA Jetson series) and investigate model compression techniques—such as structured pruning and weight quantization—to maintain real-time performance on resource-constrained hardware. Additionally, broadening the model's scope to detect diverse fruit types under various environmental conditions will enhance scalability and practical applicability. Integrating this model with IoT-based agricultural management systems may also provide seamless real-time monitoring, enabling precision agriculture practices such as targeted fertilizer, pesticide, and water applications. Ultimately, this could lead to more sustainable farming methods and reduced environmental impact.

In summary, this study establishes a solid foundation for continued development in automated agricultural technologies, highlighting significant potential for future enhancements and broad-scale adoption in global agriculture.

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